Light

GCSE Computer Science — Programming Testing and Robustness (Set 3)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What happens if a syntax error is not corrected before running a program?

- ○ **A.** The program runs but produces incorrect output
- ○ **B.** The program will not run at all
- ○ **C.** The program runs slower than expected
- ○ **D.** The program automatically fixes the error

Q2 (1 mark)

A program should grant access if a user is an admin OR has permission. The code incorrectly uses AND instead of OR. What type of error is this?

- ○ **A.** Syntax error
- ○ **B.** Logic error
- ○ **C.** Runtime error
- ○ **D.** Type error

Q3 (1 mark)

A function calls itself repeatedly without ever stopping. What type of error will eventually occur?

- ○ **A.** Syntax error
- ○ **B.** Logic error only
- ○ **C.** Runtime error (stack overflow)
- ○ **D.** No error will occur

Q4 (1 mark)

What is the purpose of recording expected outputs in a test plan?

- ○ **A.** To compare actual results against what the program should produce
- ○ **B.** To make the program run faster
- ○ **C.** To reduce the number of tests needed
- ○ **D.** To automatically generate test data

Q5 (1 mark)

Which columns should a complete test table include?

- ○ **A.** Input data only
- ○ **B.** Test number, input data, expected output, actual output, pass/fail
- ○ **C.** Expected output and actual output only
- ○ **D.** Test number and pass/fail only

Q6 (1 mark)

What is a check digit?

- ○ **A.** A password requirement for minimum digits
- ○ **B.** The first digit of a number that identifies its type
- ○ **C.** A digit that must be entered twice for confirmation
- ○ **D.** An extra digit calculated from other digits to detect errors

Q7 (1 mark)

What is a lookup table validation?

- ○ **A.** Creating a table of all possible errors
- ○ **B.** Checking if entered data exists in a predefined list of valid values
- ○ **C.** Looking up the user's password in a database

- ○ **D.** Checking if a table exists in the database

Q8 (1 mark)

What is a breakpoint in debugging?

- ○ **A.** A marker that pauses program execution at a specific line
- ○ **B.** A point where the code is broken and won't compile
- ○ **C.** The location where an error message is displayed
- ○ **D.** A command that stops the program permanently

Q9 (1 mark)

Why is version control important for program development?

- ○ **A.** It makes the program execute faster
- ○ **B.** It reduces the memory the program uses
- ○ **C.** It prevents syntax errors from occurring
- ○ **D.** It tracks changes and allows reverting to previous versions

Q10 (1 mark)

What is the purpose of technical documentation for a program?

- ○ **A.** To explain how the program works for future maintenance

- B. To make the program file smaller
- C. To prevent unauthorised users from running the program
- D. To increase the program's execution speed

Q11 (1 mark)

What is two-factor authentication (2FA)?

- A. Entering a password twice
- B. Using two different passwords
- C. Verifying identity using two different types of evidence
- D. Having two users approve access

Q12 (1 mark)

What is a dry run in program testing?

- A. Running the program without any input data
- B. Testing the program in a desert environment
- C. Running the program with the monitor turned off
- D. Manually tracing through code on paper without executing it

Q13 (2 marks)

Consider the following Python code:

```
def greet(name):
print("Hello", name)
```

What type of error does this code contain?

- ○ **A.** Logic error
- ○ **B.** Runtime error
- ○ **C.** Syntax error (indentation)
- ○ **D.** No error

Q14 (2 marks)

A program should print all items in a 5-element list. Consider the code:

```
items = ["a", "b", "c", "d", "e"]
for i in range(1, 5):
    print(items[i])
```

The program misses one item. What is the error?

- ○ **A.** Runtime error: index out of range
- ○ **B.** Syntax error: incorrect range usage
- ○ **C.** Logic error: should be range(0, 5)
- ○ **D.** Logic error: should be range(1, 6)

Q15 (2 marks)

What is the purpose of double entry validation?

- ○ **A.** To check if data is within a valid range
- ○ **B.** To require users to enter critical data twice and compare
- ○ **C.** To enter data into two different databases
- ○ **D.** To check data against a lookup table twice

Q16 (2 marks)

What does 'stepping through code' mean in debugging?

- ○ **A.** Deleting lines of code one at a time
- ○ **B.** Jumping to specific functions in the code
- ○ **C.** Running the program multiple times
- ○ **D.** Executing one line at a time and examining the results

Q17 (2 marks)

What is regression testing?

- ○ **A.** Retesting to ensure changes haven't broken existing features
- ○ **B.** Testing that the program runs backwards
- ○ **C.** Testing with progressively smaller data sets
- ○ **D.** Testing only the newest features

Q18 (2 marks)

What does graceful degradation mean in software?

- ○ **A.** The program gradually becomes slower over time
- ○ **B.** The program displays polite error messages
- ○ **C.** The program automatically deletes old files
- ○ **D.** The program continues working with reduced functionality when errors occur

Q19 (2 marks)

Consider the following code:

```
age = int(input("Enter your age: "))
```

What type of error occurs if the user enters "eighteen"?

- ○ **A.** Syntax error
- ○ **B.** Logic error
- ○ **C.** Runtime error (ValueError)
- ○ **D.** No error occurs

Q20 (2 marks)

Which variable naming convention is commonly used in Python?

- ○ **A.** ALLCAPS (e.g., TOTALVALUE)

- ○ **B.** kebab-case (e.g., total-value)
- ○ **C.** snake_case (e.g., total_value)
- ○ **D.** No convention exists

Q21 (2 marks)

What is unit testing?

- ○ **A.** Testing individual functions or components in isolation
- ○ **B.** Testing the entire program as one unit
- ○ **C.** Testing with only one piece of test data
- ○ **D.** Testing performed by one person

Q22 (2 marks)

Which is an example of biometric authentication?

- ○ **A.** Entering a PIN code
- ○ **B.** Answering security questions
- ○ **C.** Inserting a smart card
- ○ **D.** Using a fingerprint scanner

Q23 (2 marks)

When is final testing (terminal testing) performed?

- ○ **A.** Continuously throughout development
- ○ **B.** Before writing any code
- ○ **C.** Only when the complete program is finished
- ○ **D.** Only by end users

Q24 (3 marks)

A program calculates 15% discount on prices. Which test table entry is correct for testing with price £80?

- ○ **A.** Input: £80, Expected: £12, Actual: £12, Pass
- ○ **B.** Input: £80, Expected: £92, Actual: £92, Pass
- ○ **C.** Input: £80, Expected: £80, Actual: £80, Pass
- ○ **D.** Input: £80, Expected: £68, Actual: £68, Pass

Q25 (3 marks)

The following code should calculate the product of numbers 1 to 4 (which is 24), but outputs 0:

```
result = 0
for i in range(1, 5):
    result = result * i
print(result)
```

What is the error and how should it be fixed?

- ○ **A.** Logic error: change 'result = 0' to 'result = 1'

- B. Syntax error: change '*' to '+'
- C. Runtime error: change 'range(1, 5)' to 'range(0, 4)'
- D. Logic error: change 'result * i' to 'result + i'

Q26 (3 marks)

Which validation checks should be applied to an email address field?

- A. Range check only
- B. Length check only
- C. Presence check and format check (contains @ and .)
- D. Type check only

Q27 (3 marks)

Consider the following code:

```
n = 20
while n >= 5:
    n = n - 3
print(n)
```

What value is output?

- A. 5
- B. 2
- C. 6

- D. -1

Q28 (3 marks)

What is a fail-safe default in program design?

- A. A backup power supply for the computer
- B. A default setting that maintains security when errors occur
- C. A safe place to store backup files
- D. A message displayed when the program crashes

Q29 (3 marks)

What is the purpose of integration testing?

- A. Testing individual functions in isolation
- B. Testing the program's user interface
- C. Testing with integrated development environments
- D. Testing how different modules work together

Q30 (3 marks)

Consider this input validation code:

```
def safe_divide(a, b):
    if b == 0:
```

```
        return "Error: Cannot divide by zero"
    return a / b
```

What robustness technique does this demonstrate?

- ○ **A.** Type checking before operations
- ○ **B.** Encryption of sensitive data
- ○ **C.** Input bounds checking to prevent errors
- ○ **D.** Compression of return values

Total: **55** marks • Time: **30 min**

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